

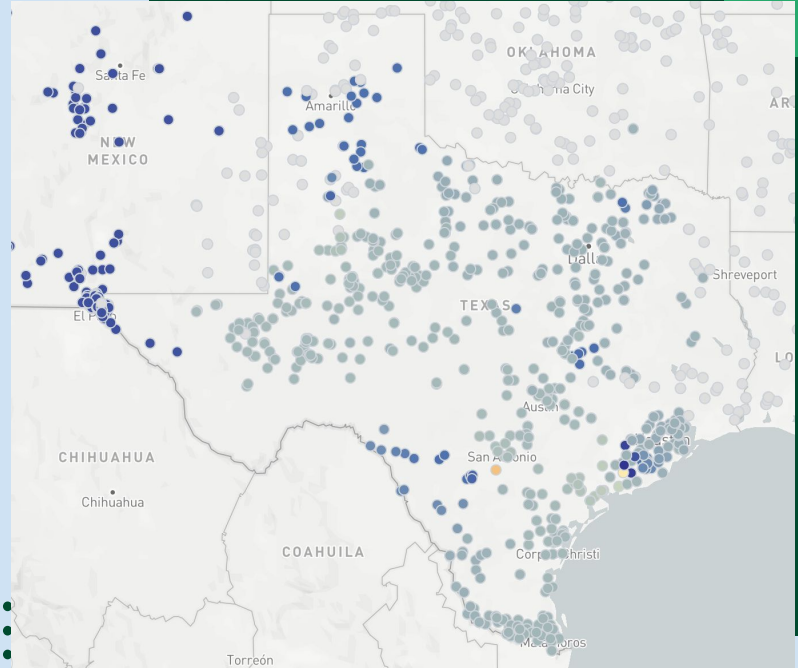
ERCOT Load Forecasting

Using load data from 2017-2025

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[Link to github:](https://github.com/alex-lopez70/Chuang)

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WhiteheadLopez_ENV797_TSA_FinalProject.git



Background and Motivation

Load growth used to be fairly stagnant and was managed by implementing more efficiency measures (Walton, 2023). But with innovation in electrical technologies (EVs, heat pumps, etc) and introductions of new load (crypto and data centers) electrical load has shot up in recent years, and this means we need new measures of responding to this load growth like demand response, interconnection reform, among others.

The ability to forecast load growth is interesting because there are both general trends (growth in electric technologies, climate change) and seasonal trends that impact load.

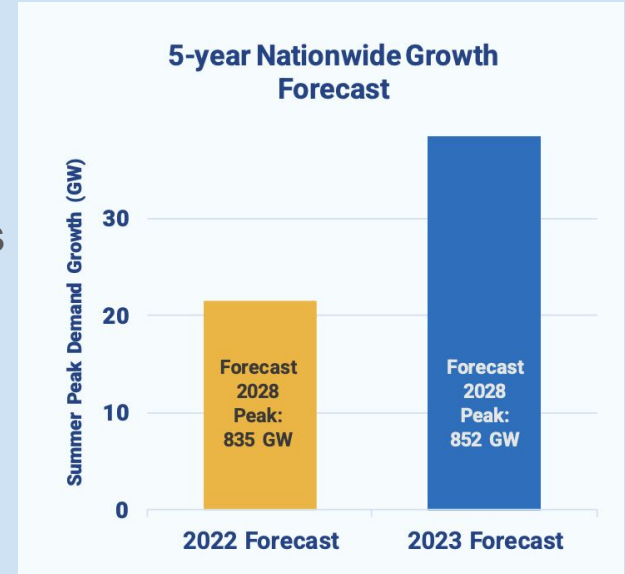


Fig. 1 Nationwide forecasts for electrical demand rose from 2.6% to 4.7% growth over the next five years, as of 2023. This corresponds to a growth of 38 GW by 2028. (Zimmerman and Wilson, 2023)

Focusing on ERCOT

ERCOT is a market witnessing tremendous load growth. In 2018, summer peak load record was 69.5 GW, but in just 5 years, this record was broken with a 16 GW increase with the new record hitting 85.6 GW (Wilson and Zimmerman, 2023). Moreover, ERCOT set a new winter demand peak in February 2025 with load being above 80 GW for the first time, similar figures usually seen in summer months, which highlights load growth not being exclusive to the summer (Kleckner, 2025). A lot of this can be attributed to crypto mining facilities, data centers, and industrial electrification (Guo, 2025).

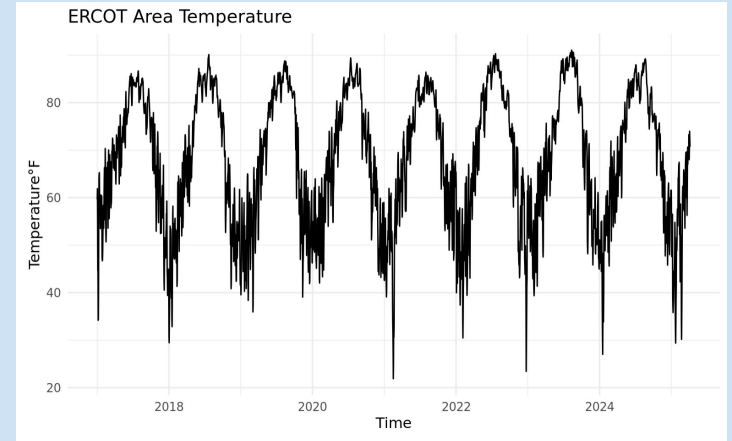
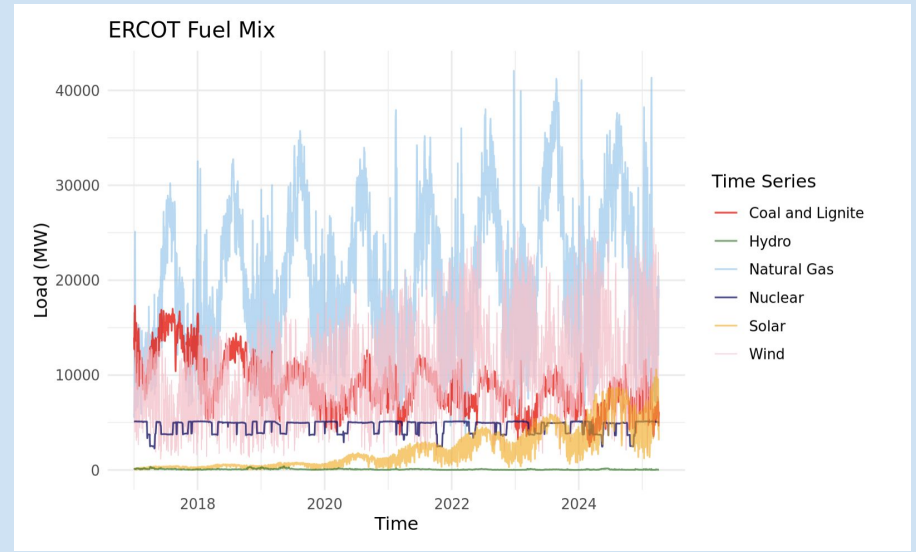
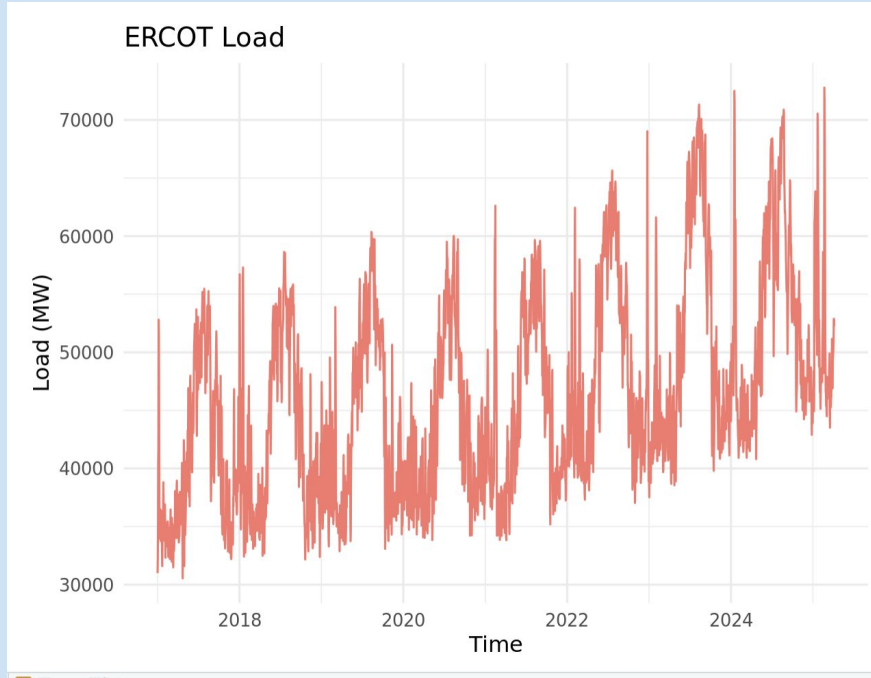
At the same time, ERCOT has had some really exciting developments, from being the wholesale market with massive growth in solar and storage, to having the fastest interconnection process, which have all helped the market respond to load growth in the wake of extreme weather events like Winter Storm Uri in 2021. Being able to better forecast load growth will help push for more policy and technology to support load growth while continuing to reliably serve pre-existing load.

Data Set Description

Source of Data: GridStatus.io and ERCOT

	Load	Temperature	Fuel Mix
Units	MW	Fahrenheit	MW
Frequency	Daily	Daily	Daily
Subsets		ERCOT Zone	Fuel (Coal and lignite, Hydro, Nuclear, Solar, Wind, Natural Gas)

Data Exploration

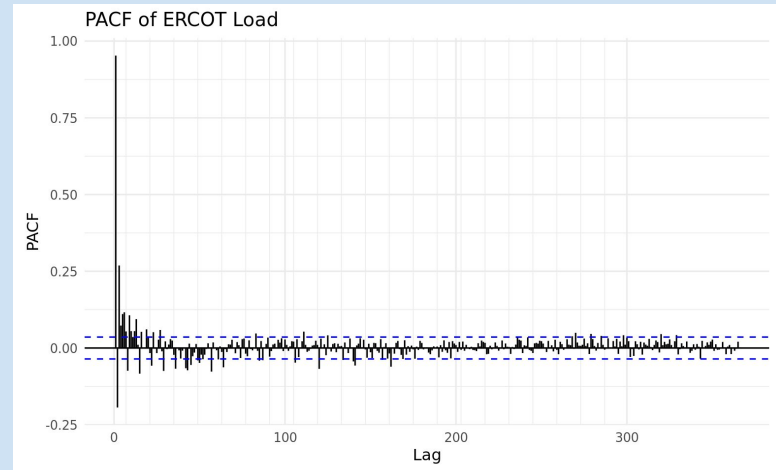
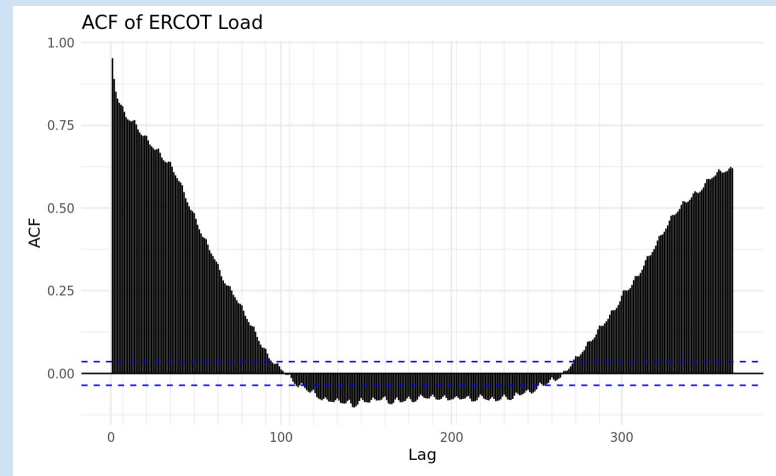


Data Exploration Cont.

A slowly decaying ACF and a choppy PACF indicate a non-stationary time series with a possible trend and seasonality.

The slow decay in the ACF suggests that past values are strongly correlated with future values.

The PACF, with significant spikes at multiple lags, can point to a seasonal pattern.



Methodology

Data Cleaning

- Pulled 2017-2025 daily data for load, temperature, and fuel mix
- Temperature forecasts divided by zone → found average each hour for simplification

Time Series Creation

- Created multi-seasonal time objects for our predictor and response variables to represent complex seasonalities in load data
 - `seasonal.periods = c(7,365.25)`
 - `start = c(2017,1,1)`
- Created train/test split
 - Test on final 365 days

Methodology - Model Creation

STL+ETS: Combines time series decomposition and exponential smoothing

TBATS: Trigonometric seasonality + Box-Cox + ARIMA + Trend + Seasonal

ARIMA + Fourier: Uses Fourier terms as exogenous regressors to model seasonal components

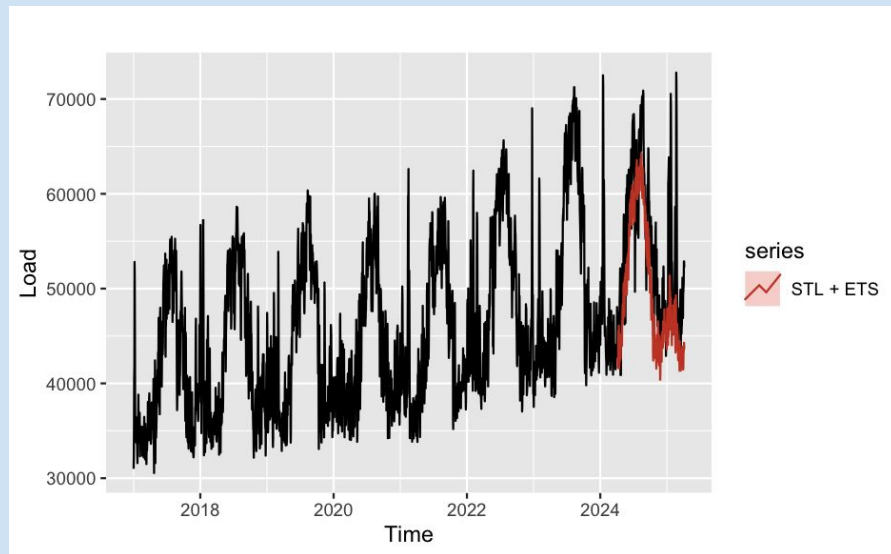
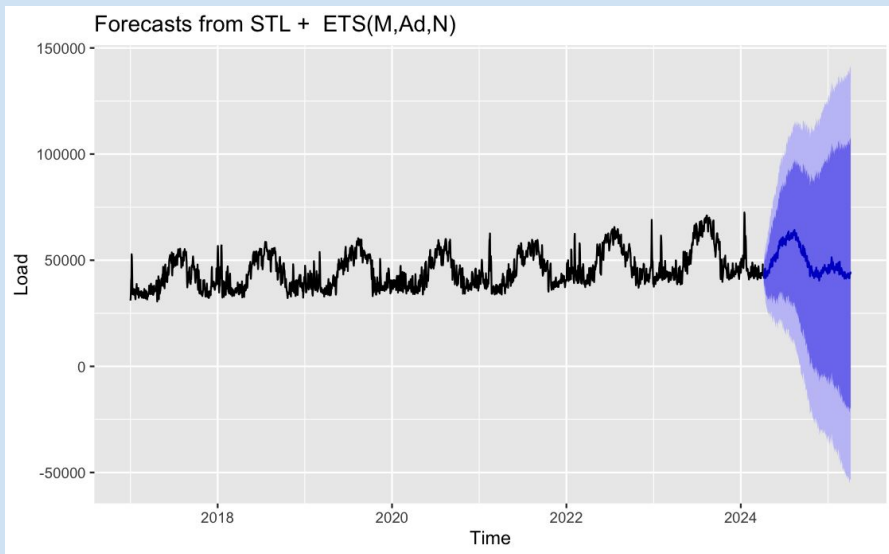
Neural Network: Uses lagged values of the series as inputs and Fourier terms to model multiple seasonalities

SARIMA: Autofits model by capturing single seasonality

SSES: Probabilistic state space model that autofits trend and seasonality

SSBSM: Probabilistic state space model that assumes each component is stochastic

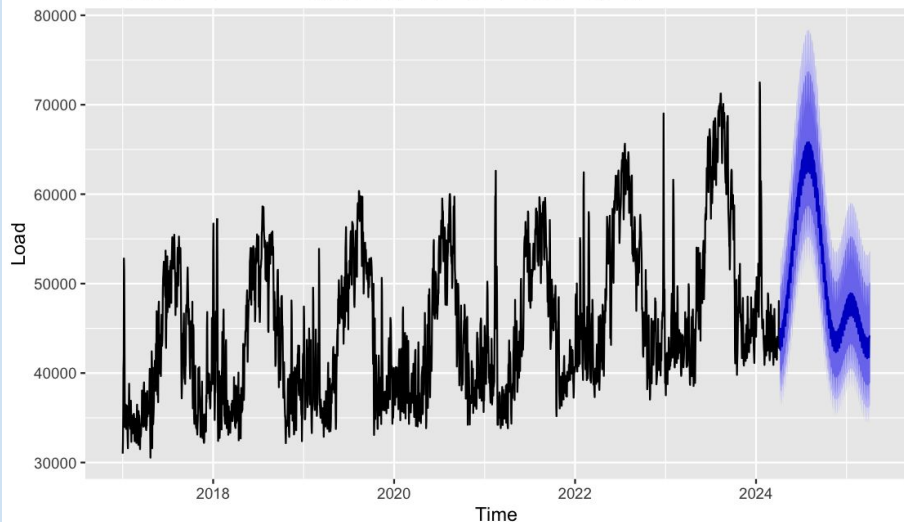
STL+ETS



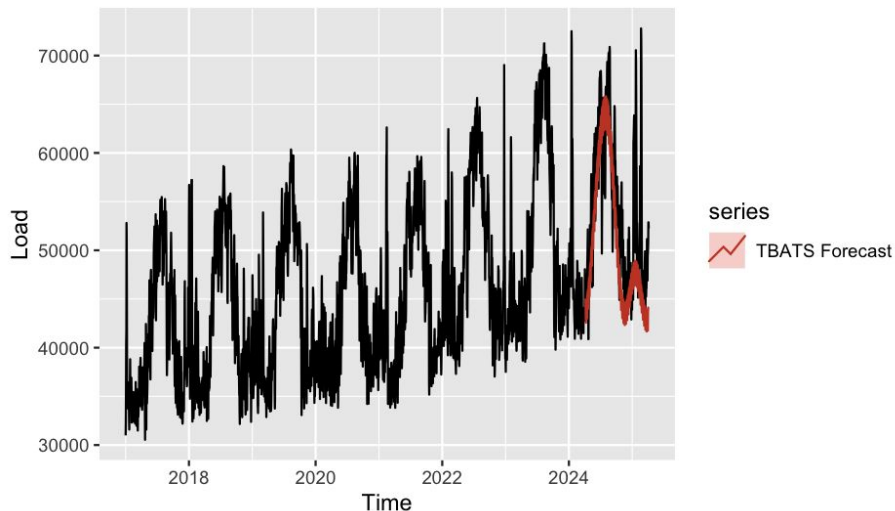
- Left plot: historical load and forecast horizon, uncertainty increases further out in time
 - Noted that prediction confidence interval lower bound dips below 0
- Model captures repeating seasonal patterns, but variance looks to grow rapidly
- STL+ETS can capture seasonal peaks and valleys, but underestimates recent volatility

TBATS

Forecasts from TBATS(0, {2,1}, -, {<7,3>, <365.25,3>})

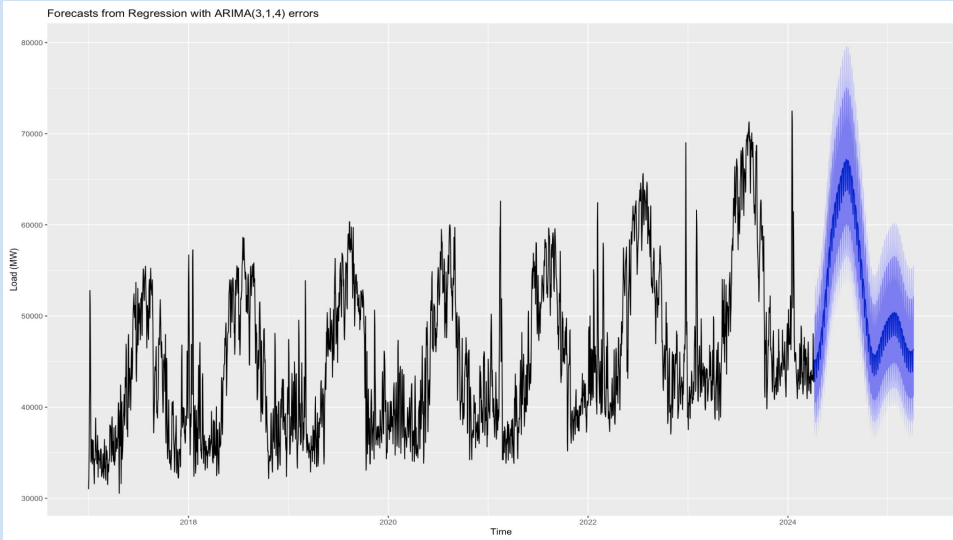


Observed Data and TBATS Forecast



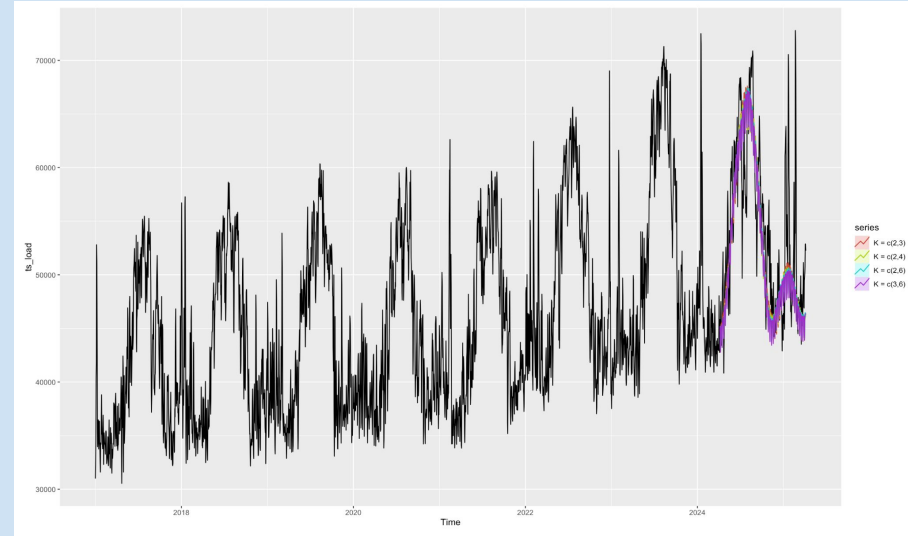
- The model seems to capture seasonal patterns and has changing amplitude over time
- Forecast aligns with more recent trends, but does under-predict stronger peaks
- It smooths out some sharp peaks and outliers

ARIMA + Fourier Terms

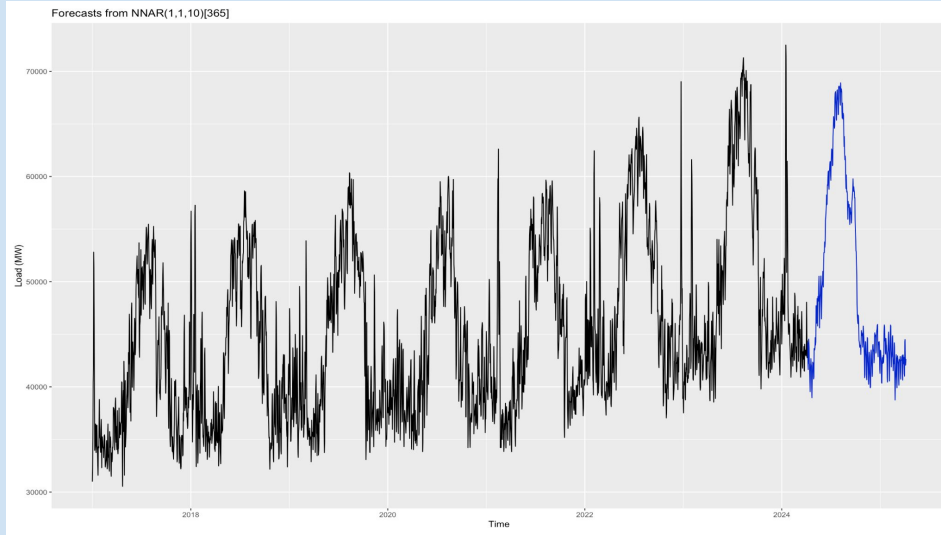


Model with K parameter $K = c(3,6)$

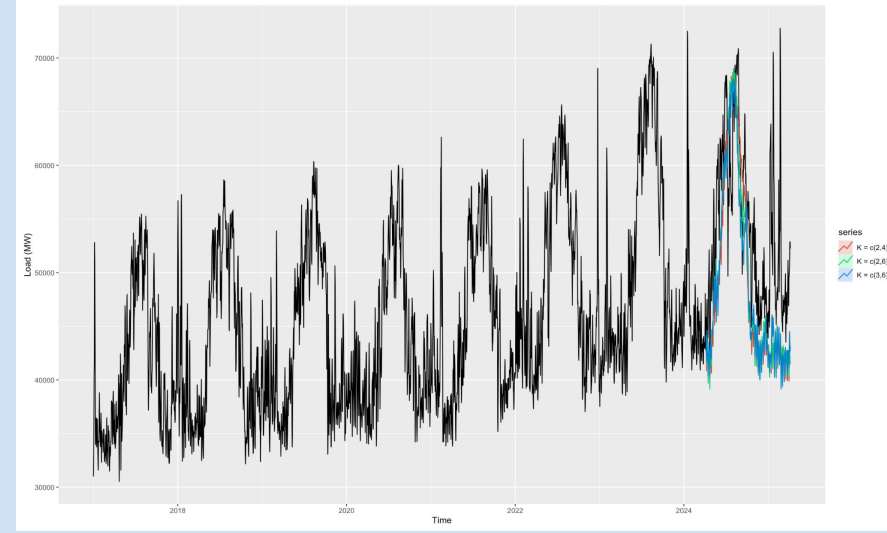
- Multiple seasonalities
- Variety of K parameters



Neural Network

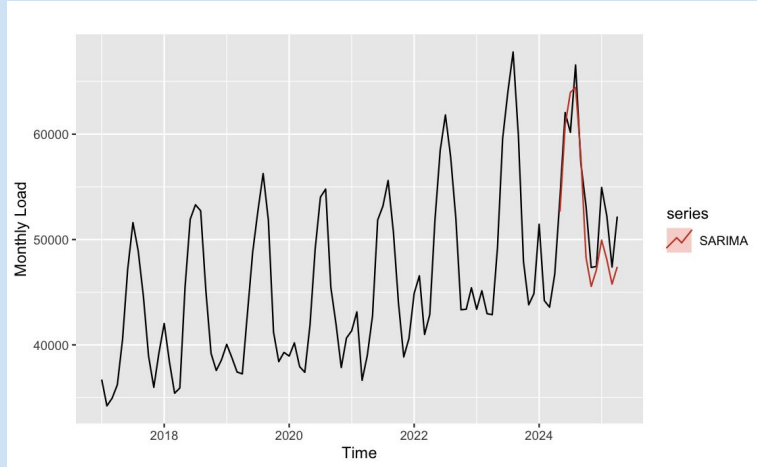


Model with K parameter $K = c(3,6)$



- Addition of p and P captures short- and long-term dependencies
- Analogous to SARIMA model

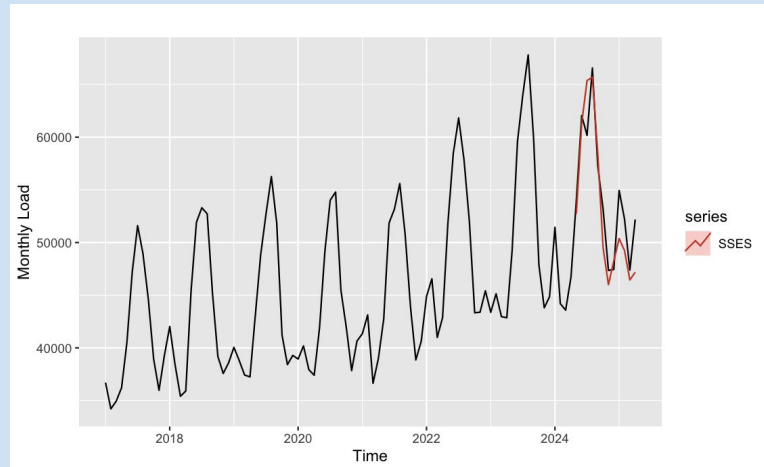
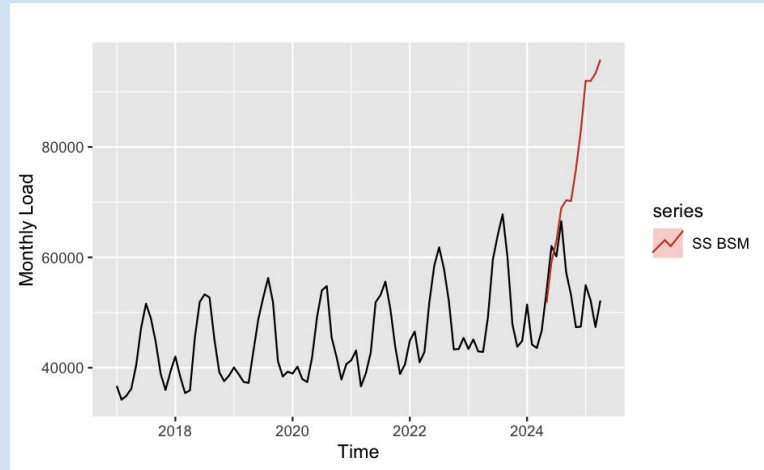
Monthly Models



SARIMA: closely follows the test data

SS BSM: Forecast explodes after 2024, greedily follows recent trends

SSSES: Similar to SARIMA, assigns weight to recent observations while still considering past values

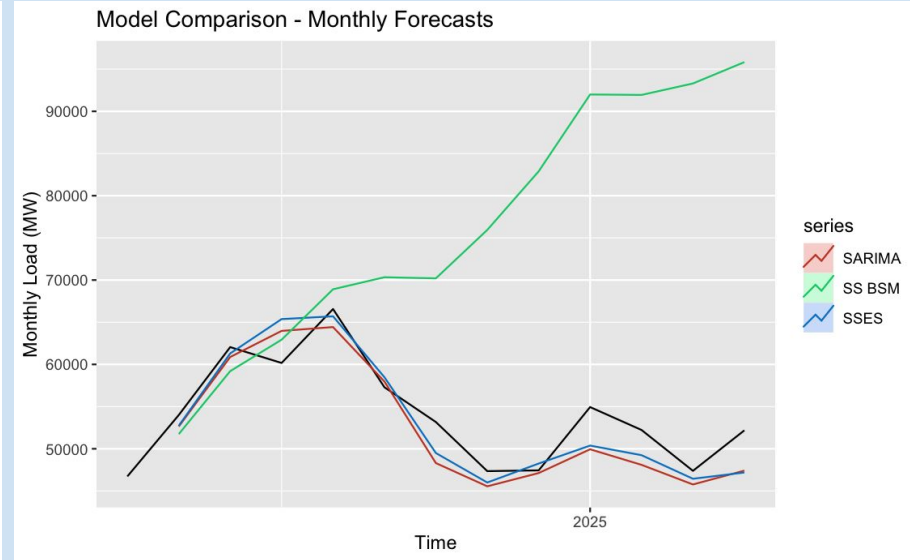
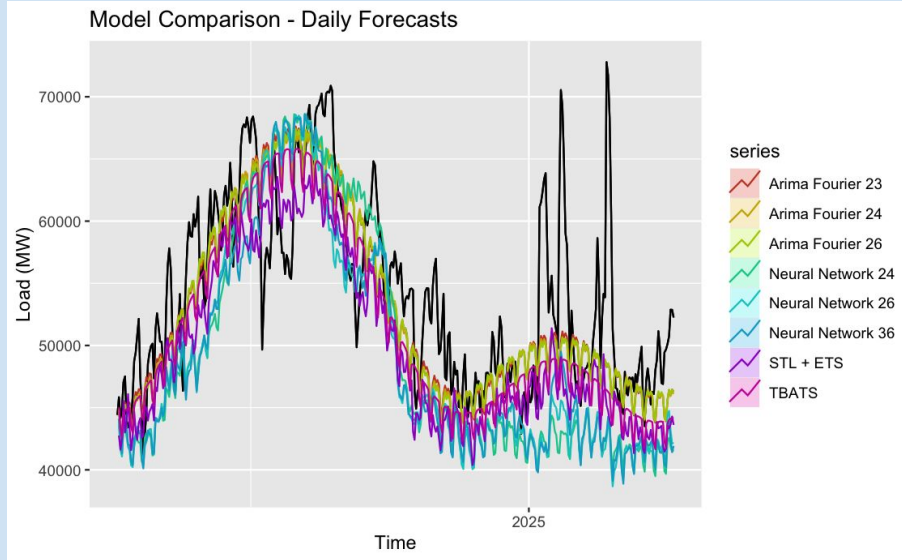


Results

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
NN24	4545.538	7567.275	6135.337	8.39830	11.20964	0.90500	2.60672
NN26	4927.829	7775.475	6286.919	9.02852	11.42744	0.89835	2.66283
NN36	5282.429	7600.326	6117.556	9.58254	11.09920	0.87819	2.59071
SARIMA	1891.987	3125.671	2638.377	3.61048	4.86036	-0.06288	0.59759
SSES	1183.083	2929.559	2383.270	2.31508	4.38636	-0.02696	0.55750
SSBM	-21702.918	28172.579	22554.935	-42.77079	44.23444	0.81247	5.64407
STL+ETS	4320.595	6453.940	4992.943	7.57688	8.85220	0.83777	2.13999
TBATS	2905.599	5794.056	4369.840	5.03801	7.80851	0.86071	1.94380
AF23	1458.268	5148.588	3746.586	2.31101	6.68713	0.85424	1.72536
AF24	1521.098	5166.958	3756.300	2.42856	6.70843	0.85414	1.73282
AF26	1545.030	5166.779	3745.856	2.47411	6.68810	0.85374	1.73143
AF36	1514.696	5155.774	3732.166	2.41897	6.66264	0.85621	1.72704

*Note: SARIMA, SSES, and SSBM forecast monthly instead of daily values

Results



- Actual test load data shown in black
- Models fail to capture large swings in daily load
- Better performance (visually) when forecasting monthly averages
 - Though SS BSM appears to diverge strongly from what is expected, suggesting that it may be extrapolating a specific recent trend too aggressively

Discussion

Best Models: ARIMA + Fourier with $K = c(2,3)$
and $K = c(3,6)$

Good Model: TBATS

Worst Model: SSBSM

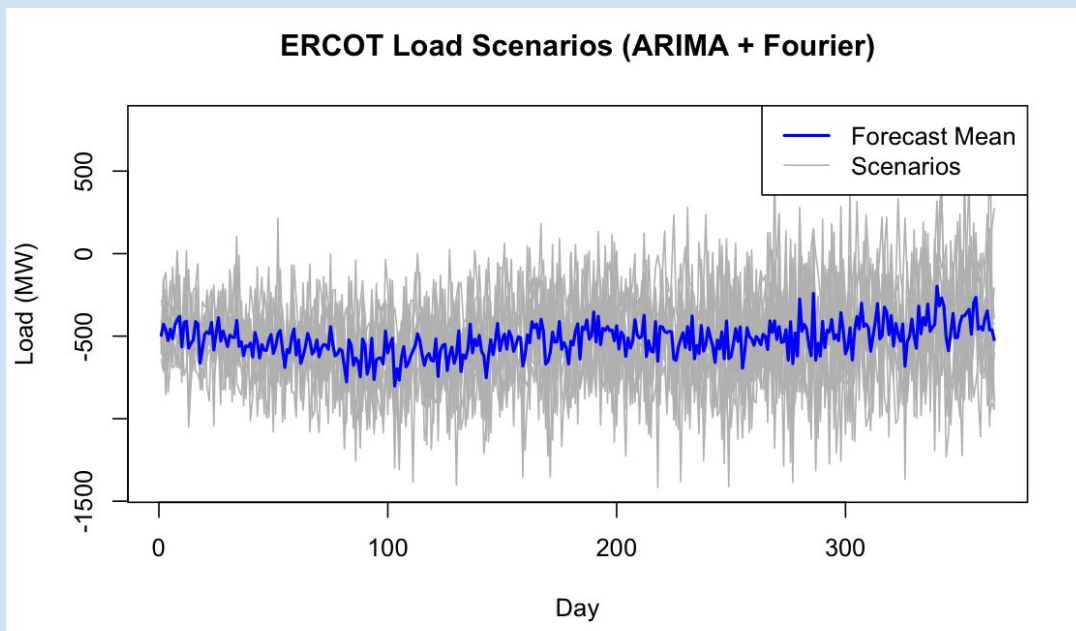
Conclusions

Daily forecasting benefits from more flexibility like ARIMA Fourier, and non-linear models like Neural Networks.

Simpler and stable models, like SARIMA and SSES, are helpful for generating monthly forecasts in this scenario as they seem to generalize better.

Next Steps

Figure out generating scenarios with varying impacts from fuel mix and temperature. Our results currently don't make sense, but it partially may be tied to load volatility!



Citations

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